Activity 5 - Build the GHG Source, Sink and Scope Register

Alignment: K4 | K5 | A2 | A7 | LO2 Duration: 45 minutes Objective: Identify gases, sources, sinks and reservoirs; classify them consistently and assess significant indirect emissions.

Scenario

Potential sources include natural-gas curing ovens, emergency diesel, company vans, refrigerant leakage, purchased electricity, rooftop solar, aluminium and electronics, inbound freight, outsourced warehousing, business travel, employee commuting, product electricity use and end-of-life treatment. Facilities also claims landscaped trees and purchased credits reduce the inventory total.

Materials

- Operational and value-chain process narrative
- Energy, maintenance, procurement and logistics summaries
- Scope and ISO category prompts
- Significance and ownership workbook

Questions

- Which gases and physical sources are present?
- How should each item be classified?
- Which indirect sources are significant and why?
- What evidence is needed for claimed removals or credits?

Detailed procedure

1. Walk the organisational and product process maps and list every plausible source, sink and reservoir.
2. Identify relevant gases for combustion, refrigerants, purchased energy, materials, transport, waste and product use.
3. Classify each source into Scope 1, 2 or 3 and the applicable ISO direct or indirect-emission category.
4. Keep biogenic emissions, removals, avoided emissions and purchased credits separately identified.
5. Define significance criteria using magnitude, influence, risk, stakeholder interest, data quality and reduction opportunity.
6. Score indirect sources and record evidence and judgement supporting each score.
7. Select sources for inclusion and document transparent reasons for exclusions or provisional estimates.
8. Assign data owner, source record, collection frequency, control and reviewer to each included source.
9. Run a completeness challenge against ledger, maintenance, procurement and waste records.

Required evidence

- Source, sink and reservoir register
- Scope and ISO category map

- Significance assessment
- Data-owner and completeness matrix

Acceptance check

All scenario sources and gases are addressed; Scope and ISO categories are consistent; removals and credits remain separate; significance and ownership decisions are evidence-based.

Files

- Activity-05-Workbook.xlsx - editable working file
- Activity-05-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.